UNIT II

(1)

CELLULAR ARCHITECTURE

Multiple Access Techniques :-

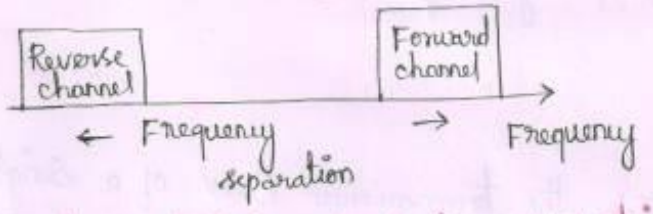
- In order to accomodate many mobile users within the finite amount of radio spectrum the multiple access techniques are used.

Three types of duplexing techniques are used. They are

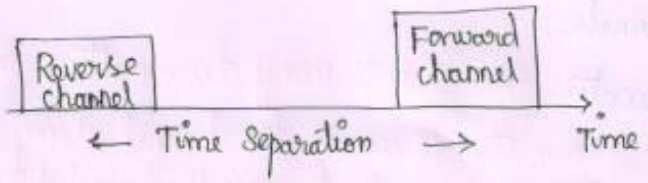
1. Simplex
2. Half duplex
3. Full duplex

The duplexing may be done by using frequency or time domain techniques.

FDD: The frequency division duplexing provides two distinct bands of frequencies for every user.



TDD: The time division duplexing uses time instead of frequency to provide a forward and reverse link.



Types of Multiple access Techniques

3 major access techniques are used in wireless communication system. The types are

1. Frequency Division Multiple Access (FDMA)
2. Time Division Multiple Access (TDMA)
3. Code Division Multiple Access (CDMA)

- These technique can be grouped as narrowband & wideband systems, depending upon how the available bandwidth is allocated to the users.

Narrow band Systems :-

- In a narrowband multiple access system, the available radio spectrum is divided into a large no. of narrowband channels. The channels are operated by using FDD.

- The interference b/w fwd & reverse links on each channel is minimize by frequency separation is made within the frequency spectrum.

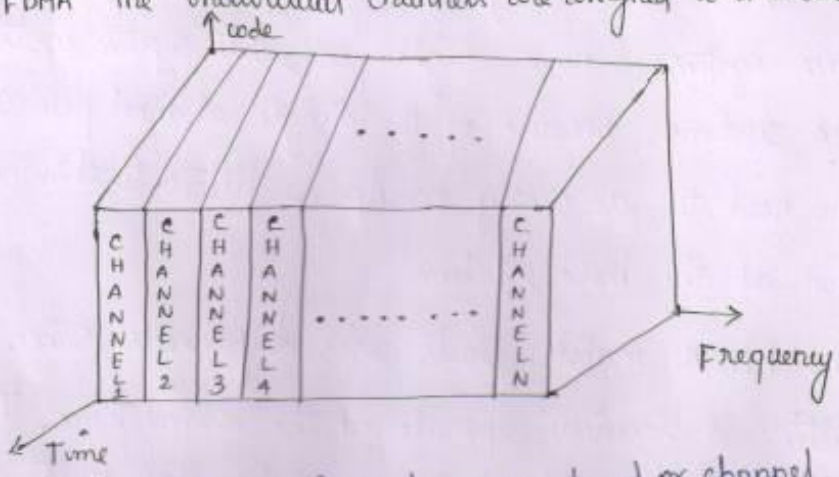
Wideband Systems :-

- In this system the transmission B.W of a single channel is much larger than the coherence B.W of the channel. In this system a large number of transmitters are allowed to transmit on the same channels.

- TDMA allocates time slots to many transmitters on the same channel & allows only one transmitter to access the channel at any instant of time but in CDMA it allows all of the transmitter to access the channel at the same time. The

Frequency Division Multiple Access (FDMA)

In FDMA the individual channels are assigned to a individual users.



- Each user is allocated a unique frequency band or channel.
- These channel are assigned on demand to users who request the service. During the period of the call, no other user can share the same channel.
- In FDMA/FDD systems, the users are assigned a channel as a pair of frequencies one frequency is used for the forward channel, while the other frequency is used for the reverse channel.

Features of FDMA:

1) The FDMA channel carriers only one phone ciruit at a time
2) The FDMA is usually implemented in narrowband systems.
3) If an FDMA channel is not in use, then it sits idle & cannot be used by other users to ↑ or share capacity. It is essentially a wasted resource.
4) After the assignment of a voice channel, the base station & the mobile transmit simultaneously & continuously.
5) The complexity of FDMA mobile systems is lower when

6) FDMA is a continuous transmission scheme, fewer bits are needed for overhead proposes as compared to TDMA.

7) FDMA systems have higher cell site system costs as compared to TDMA systems, because of the single channel per carrier design and the need to use costly bandpass filters to eliminate spurious radiation at the base station

8) The FDMA mobile units uses duplexers since both the transmitter & receiver operate at the same time. This results in an ↑ in the cost of FDMA subscriber units & base stations

9) FDMA requires tight RF filtering to minimize adjacent channel interference.

Non linear effects of FDMA:-

- In a FDMA system, many channels share the same antenna at the base station. The power amplifiers or the power combiners when operated at or near saturation for maximum power efficiency are non-linear.

- The non-linearities causes signal spreading in the frequency domain & generate intermodulation frequencies. IM can interfere with other channels in the FDMA systems.

- IM is the generation of undesirable harmonics. The no. of channels than can be simultaneously supported in a FDMA system is given by

$$N = \frac{B_t - 2B_{guard}}{B_c}$$

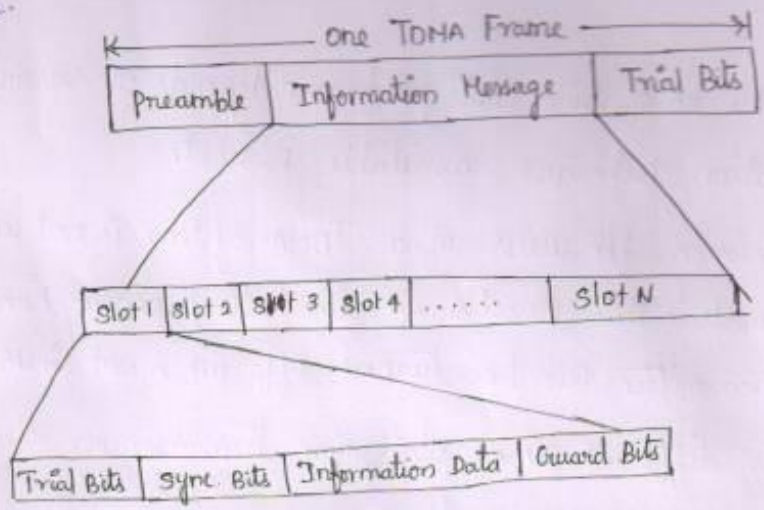
$B_t \rightarrow$ Total spectrum allocation

$B_{guard} \rightarrow$ Guard band allocated at the edge of the allocated spectrum band

$B_c \rightarrow$ channel B.W

Time Division Multiple Access

- TDMA systems divide the radio spectrum into time slots & in each slot only one user is allowed either to transmit or receive.



TDMA frame Structure

- TDMA systems transmit data in buffer & burst method. Thus the transmission for any user is noncontinuous.

- The frame consists of a no. of slots. Each frame is made up of a preamples, an information message & trial bits.

- In TDMA/TDD, half of the time slots in the frame information message would be used for the forward link channels & half used for reverse link channels.

- In TDMA/FDD, similar frame structure would be used for either forward or reverse transmission but the carrier frequency would be different for the forward & reverse links.

- In TDMA frame the preamble contains the address & synchronization information that both the base station & the subscribers used to identify each other.

- Guard times are utilized to allow synchronization of the receivers for different slots & frames. Different TDMA wireless standards have different TDMA frame.

Features of TDMA

1) In TDMA the no. of time slots per frame depends on several factors such as modulation technique, available B.W etc.

2) Data transmission for users of a TDMA system is not continuous but occurs in bursts. This results in slow battery consumptions, since the subscriber transmitter can be turned off when not in use.

3) TDMA uses different time slots for transmission & reception thus duplexers are not required.

4) Adaptive equalization is usually necessary in TDMA systems, since the transmission rates are generally very high as compared to FDMA channels

5) In TDMA the guard time should be minimized.

6) High synchronization overhead is required in TDMA systems because of burst transmissions

7) Advantage: It is possible to allocate different number of time slots / frame to different users. Thus B.W can be supplied on demand to different users by concatenating time slots based on priority

The no. of channel in TDMA system is given by

$$N = \frac{m(B_{tot} - 2B_{guard})}{B}$$

m → Maximum no. of TDMA users supported on each radio

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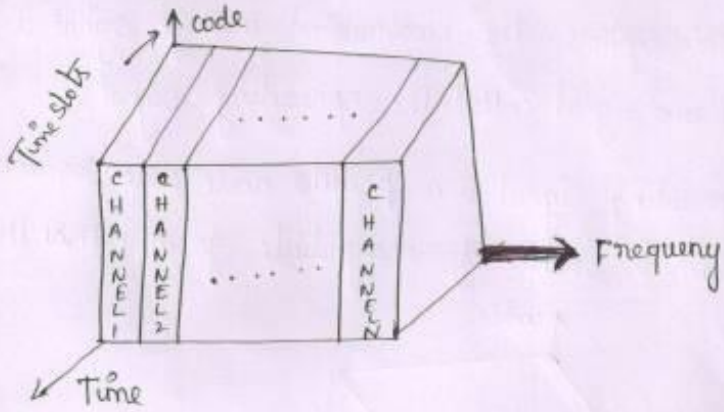
TDMA scheme



Efficiency of TDMA:-

- The efficiency of TDMA system is a measure of the percentage of transmitted data that contains information as opposed to providing overhead for the access scheme.

The frame efficiency η_f is the percentage of bits / frame which contains transmitted data.

The no. of overhead bits / frame is

$$b_{OH} = N_r b_r + N_t b_p + N_t b_g + N_r b_g$$

$N_r \rightarrow$ No. of reference bursts / frame

$N_t \rightarrow$ No. of traffic burst / frame

$b_r \rightarrow$ No. of overhead bits / reference burst

$b_p \rightarrow$ No. of overhead bits / preamble in each slot

$b_g \rightarrow$ No. of equivalent bits in each guard time interval

The total no. of bits / frame is

$$b_T = T_f R$$

T_f - frame duration

R - channel bit rate

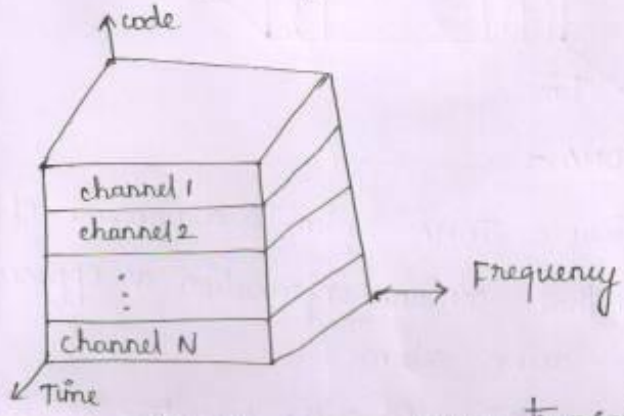
The frame efficiency η_f is given as

$$\eta_f = \left(1 - \frac{b_{OH}}{\ }\right) \times 100\%$$

Code Division Multiple Access

- In this systems, the narrowband message signal is multiplied by a very large b.w signal called the spreading signal.

- The spreading signal is a pseudo noise code sequence that has a chip rate which is orders of magnitude greater than the data rate of the message.



- Uses the same carrier frequency as may transmit simultaneously

- Each user has its own pseudorandom codeword which is approximately orthogonal to all other codewords.

- Each user operates independently with no knowledge of the other users.

- The near far problem occurs when many mobile users share the same channel.

- In general the strongest received mobile signal will capture the demodulator at a base station. To combat the near-far problem power control is used in most CDMA implementation

- power control is implemented at the base station by rapidly sampling the radio signal strength indicator levels of each mobile as then sending a power change command over the forward radio link.

(5)

Features of CDMA

1) The CDMA system many users share the same frequency either TDD or FDD may be used

2) Unlike TDMA or FDMA, CDMA has a soft capacity limit

3) Multipath fading may be substantially reduced because the signal is spread over a large spectrum.

4) channel data rates are very high in CDMA systems

5) CDMA uses co-channel cells it can use macroscopic spatial diversity to provide soft handoff. Soft handoff is performed by the MSC (i.e) The MSC can simultaneously monitor a particular user from 2 or more stations. Then the MSC may chose the best version of the signal at any time without switching frequencies.

6) Self jamming is a problem in CDMA system. It arises from the fact that the spreading sequences of different users are not exactly orthogonal hence in the de-spreading of a particulars PN code, non-zero contributions to the receiver decision satistic for a desired user arise from the transmission of other users in the system.

7) The near far problem occurs at a CDMA receiver if an undesired user has a high detected power as compared to the desired user.

(6)

Cellular concept

In early mobile radio system large coverage area was achieved by using a single, high powered transmitter with an antenna mounted on a tall tower, but it was impossible to reuse those same frequencies throughout the system since any attempts to achieve frequency reuse would result in interference.

The cellular concept offered very high capacity in a limited spectrum allocation without any major technological changes. It is a system level idea which calls for replacing a single high powers transmitter (large cell) with many low power transmitters (small cells) each providing coverage to only a small portion of the service area.

Frequency reuse :-

Each cellular base station is allocated a group of radio channels to be used within a small geographic area called a cell.

The design process of selecting is allocating a channel groups for all of the cellular base stations within a system is called frequency reuse (or) frequency planning

- The concept of cellular frequency reuse where cells labelled with the same letter use the same group of channels

The actual radio coverage of a cell is known as the footprint when using hexagons to model coverage areas. base stations transmitters are depicted as either being in the centre of the cell (center-excited cells) or on three of the six cell (vertices edge exited cell)

For center excited cells Omnidirectional antennas & edge exited cells sectored directional antenna are used.

To understand the frequency reuse concept consider a cellular system which has a total of S duplex channels available for use. If each cell is allocated a group of K channels (K<S) & if the S channels are divided among N cells which each have the same number of channels, the total number of available radio channels can be expressed as

$$S = KN$$

cluster:- The N cells which collectively use the complete set of frequencies is called cluster. The above figure using 3 cluster

If a cluster is repliated M times within the system the total number of duplex channels C can be used as a measure

eqn $S = KN$ the capacity of a cellular system is directly proportional to number of times a clustered is replicated in a fixed service area (7)

N - cluster size may be equal 4, 7 or 12

Frequency reuse factor of a cellular system is given by 1/N

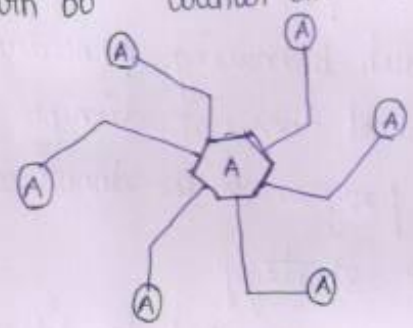
The number of cells / cluster is given by

$$N = i^2 + j^2 + ij$$

where i & j are non-negative integers. To find nearest co-channel neighbours of a particular cell, one must do the following

(i) Move i cells along any chain of hexagons

(ii) Turn 60° counter clkwise & move j cells



Channel Assignment strategies

For efficient utilization of the radio spectrum a frequency reuse is used in order to increase the capacity & minimize the interference.

The channel assignment strategies can be classified into 2 categories

1) Fixed channel assignment strategy

2) Dynamic channel assignment strategy

Fixed channel assignment strategy

- In this strategy each cell is allocated a predetermined set of voice channel.
- Any cell attempt within the cell can only be served by the unused channels in that particular cell.
- If all the channel in that cell are occupied the call is blocked & the subscriber does not receive service.
- Several variations of the fixed assignment strategy exist.
- In one approach is called borrowing strategy, a cell is allowed to borrow channels from a neighbouring cell if all of its own channels are already occupied.
- The MSC supervises such borrowing procedures & ensures that the borrowing of a channel does not disrupt or interfere with any of the calls in progress in the donar cell.

Dynamic channel assignment strategy

- Voice channels are not allocated to different cells permanently.
- Instead each time a call request is made the serving base station requests a channel from the MSC.
- Dynamic channel assignment reduce the call blocking which increases the trunking capacity of the system.

(8)

Handoff

When a mobile moves into a different cell while a conversation is in progress, the MSC automatically transfers the call to a new channel belonging to the new base station. This is called Handoff.

This handoff operation not only involves identifying a new base station but also requires that the voice & control signals be allocated to channels associated with the new base station.

Handoff must be performed successfully & as infrequently as possible & be imperceptible to the users.

The margin level at which the hand off is made is given by

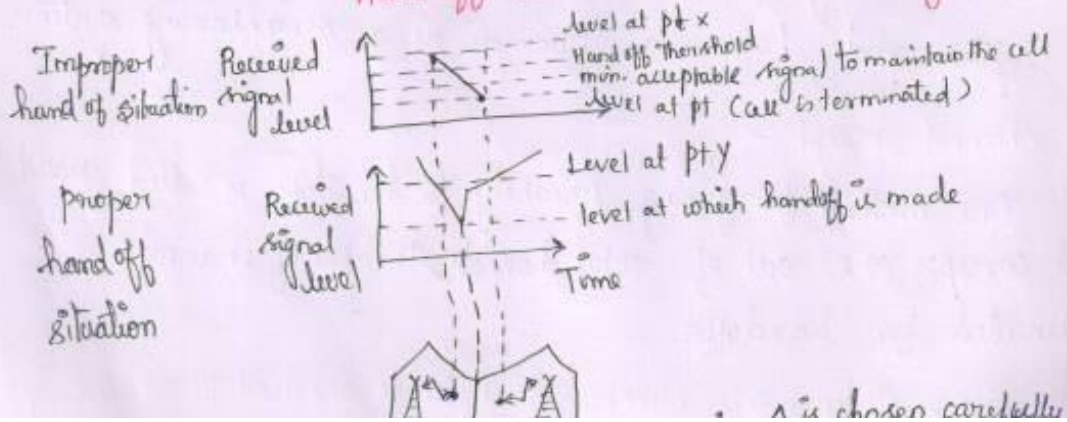
$$\Delta = P_r \text{ handoff} - P_r \text{ minimum usable}$$

$\Delta \rightarrow$ cannot be too large or too small

→ if Δ is too large, unnecessary handoffs which burden the MSC may occur

→ if Δ is too small, there may be insufficient time to complete a handoff before a call is lost due to weak signal conditions

Hand off Scenario at cell boundary

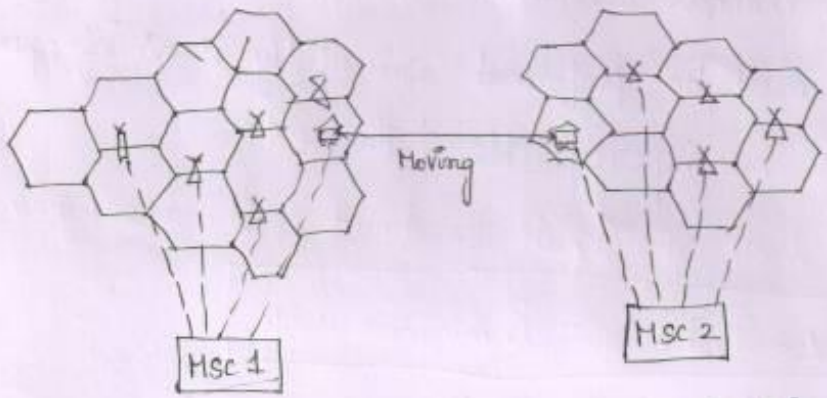


Δ is chosen carefully to meet

Inter system handoff :-

If a mobile moves from one cellular system to a different cellular system controlled by a different MSC during the process of a call a handoff is made. This type of handoff is called "intersystem handoff"

Intersystem handoff



- There are so many problems occur in this type of handoff
- when a mobile moves from one cellular system to another the MSC cannot find another cell within its system to which it can transfer the call is in progress. and also a local call may become a long distance call as the mobile moves out of its home system and becomes a roamer in a neighbouring system.

Guard channel concept :-

One method for giving priority to handoff is called guard channel concept (i.e) out of total channels, the No. of channels are allocated for handoffs.

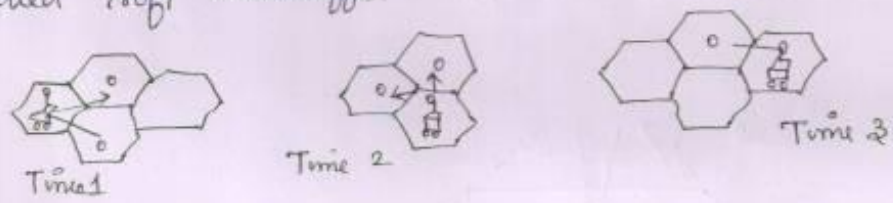
Disadvantage - Only fewer number of channels are used to Originating

Advantage:- Efficient spectrum utilization when dynamic channel assignment strategies

Dwell Time: The time over which a call may be maintained within a cell without handoff is called dwell time.

Mobile Assisted Handoff (MAHO): In this type every mobile station measures the received power from surrounding base station & continuously reports the results of these measurements to the serving base station. MAHO is particularly suited for micro cellular environment. There are 2 types of handoff's are used.

(1) Soft Handoff:- Mobile communicates with 2 or more cells at the same time & find which one is a strongest signal base station then it automatically transfers the cell to that base station is called soft handoffs.



Hard Handoff:-

If the MSC monitors the strongest signal base station & transfer the call to that base station than it is called hard handoff

(10)

- Another problem in the handoff is "cell dragging"
- It results from the pedestrain users that providing a strong signal to the base station. This type of situation occurs in the urban environment where there is a LOS radio path b/w the subscriber & the base station.
- When the user travels away from the base station at a very low speed, the average signal strength does not decay rapidly.
- Even when the user has travelled well beyond the designed range of the cell, the received signal at the base station may be above the handoff threshold. thus a handoff may not be made. This creates a potential interference & traffic management problem this is called "cell dragging"

To solve this problem handoff thersholds & radio coverage parameters must be adjusted carefully.

Interference and system capacity :-

Interference on voice channel causes cross talk where the subscribers hears interference in the background due to an undesired transmission.

There are 2 types of interference

1) Adjacent channel interference
2) co-channel interference

The co-channel reuse ratio is defined by (11)

$$Q = D/R = \sqrt{3N}$$

if Q is smaller then cluster size N also small that is providing larger capacity

Let i_0 be the number of co-channel interfering cells then the signal to interference ratio for a mobile receiver is

$$\frac{S}{I} = \frac{S}{\sum_{j=1}^{i_0} I_i}$$

$S \rightarrow$ signal power for desired base station

$I_i \rightarrow$ Interference power caused by interfering co-channel cell BS.

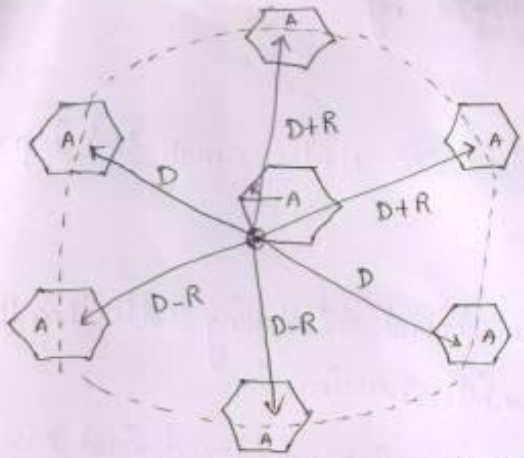
The average received power P_r at a distance d from the transmitting antenna is given by

$$P_r = P_0 (d/d_0)^{-n} \quad \text{or} \quad P_{r(dBm)} = P_{0(dBm)} - 10n \log(d/d_0)$$

When the transmit power of each BS is equal & the path loss exponent is the same throughout the coverage area, S/I for a mobile can be approximated as

$$\frac{S}{I} = \frac{R^{-n}}{\sum_{j=1}^{i_0} (D_i)^{-n}}$$

- considering only the first layer of interfering cells. if all the interfering base stations are equidistant from the desired base station & if this distance is equal to the distance D b/w cell centers then $\frac{S}{I} = \frac{(D/R)^n}{i_0} \frac{(\sqrt{3N})^n}{i_0}$



- using an exact cell geometry layout it can be shown for a 7 cell cluster with the mobile unit at the cell boundary.

- The Mobile is a distance D-R from the 2 nearest co-channel interfering cells & is exactly D+R/2. D, D-R/2 & D+R from the other interfering cells.

if n = 4

$$S/I = \frac{R^{-n}}{\sum_{j=1}^{i_0} (D_i)^{-n}} = \frac{R^{-4}}{2(D-R)^{-4} + 2(D+R)^{-4} + 2D^{-4}}$$

rewritten in terms of Q

$$\frac{S}{I} = \frac{1}{2(Q-1)^{-4} + 2(Q+1)^{-4} + 2Q^{-4}}$$

$$Q = \sqrt{3 \times N} = \sqrt{3 \times 7} = \sqrt{21} = 4.6$$

N = 7, Q = 4.6 & the worst case signal to interference ratio

$S/I \approx 49.56$

(12)

Trunking and Grade of Service :-

The GOS is a measure of the ability of a user to access a trunked system during the busiest hour. The busy hour is based upon customer demand at the busiest hour during a week, month or year.

The busy hours for cellular radio system typically occur during rush hours b/w 4 Pm & 6 pm on a Thursday (or) Friday evening

Common terms used in trunking theory

(i) Setup Time: The time required to allocated a trunked radio channel to a requesting user.

(ii) Blocked call: call which cannot be completed at a time of request, due to congestion. Also referred to as a lost call.

(iii) Traffic Intensity :- Measure of channel time utilization, which is the average channel occupancy measured is Erlangs denoted by A.

(iv) Request Rate :- The average number of call requests / unit time denoted by λ/sec the traffic Intensity $A_u = \lambda H$

λ - average no. of call request / time for each user

H - average duration of a cell

U - No. of users

For a system containing U users & an unspecified no. of channels. the total offered traffic intensity A is given as

$$A = UA_u$$

is determined by "Erlange formula" is given by

$$P_r(\text{delay}>0) = \frac{A^c}{A^c + c!\left(1-\frac{A}{c}\right)\sum_{k=0}^{c-1}\frac{A^k}{k!}}$$

- If no channels are immediately available the cell is delayed & the probability that the delayed call is forced to wait more than "t" is. The GoS of a trunked system where blocked calls are delayed is given by

$$P_r(\text{delay}>t) = P_r(\text{delay}>0)\, P_r(\text{delay}>t \mid \text{delay}>0)$$

$$= P_r(\text{delay}>0)\exp\left(-(c-A)t/H\right)$$

The average delay D for all calls in a queued system is given by

$$D = P_r(\text{delay}>0) = H/c-A$$

$H/c-A$ is the average delay for those calls which are queued.

Trunking Efficiency :-

It is a measure of the number of users which can be offered a particular GoS with a particular configuration of fixed channels.

Load: Traffic Intensity across the entire trunked radio system, measured in Erlangs.

(14)

P_r (at old cell boundary) $\propto P_{t_1} R^{-n}$ &

P_r (at new cell boundary) $\propto P_{t_2} (R/2)^{-n}$

P_{t_1} & P_{t_2} – transmit powers of the larger & smaller cell base stations

$n \rightarrow$ path loss exponent

if $n=4$ $\quad P_{t_2} = P_{t_1}/16$

∴ the cells are splitted then the transmitter power will be reduced, so that handoff's occur less frequently.

cell Sectoring

- Sectoring is the another way to ↑ the capacity of the system. with sectorization S/R is improved due to usage of directional antennas.

- Directional antennas are used to radiate power in one particular direction. The area covered by each directional antenna is called as sector so that this method is called as cell sectoring

- By using directional antennas a given cell will receive interference & transmit with only a fraction of the available co-channel cells.

- The factor by which the co-channel interference is reduced depends on the amount of sectoring used. A cell is normally partitioned into three 120° sectors or six 60° sectors

120° Sectoring

60° sectoring

- when sectoring is employed the channels used in particular cell are broken down into sectored groups and are used only within a particular sector. From that interference also ↓ & capacity ↑.

how 180° sectoring reduces interference from co-channel cells

- consider the interference experienced by a mobile located in the right most sector in the centre cell labelled "5". It will experience interference on the forward link from only two sectors out of six.

Micro cell Zone Approach

- The ↑ no. of handoffs required when sectoring is employed results in an ↑ load on the switching & control link elements of the mobile system.

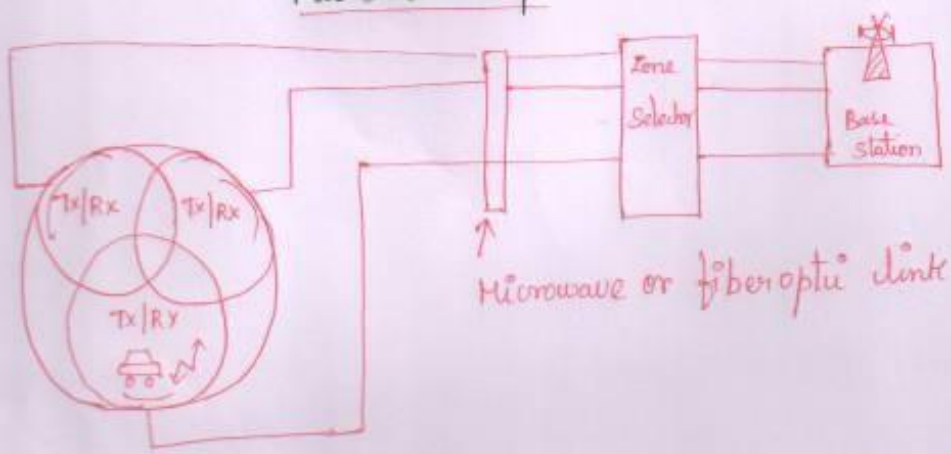
- In order to avoid the above problem micro cell zone approach is used.

(15)

- In this method cells are divided into no. of zones & each no. of zones consists of individual Transceiver.

- so that interference also reduced and capacity is ↑ because ↓ co-channel interference improves the signal quality & also leads to an ↑ in capacity without degradation in trunking efficiency caused by sectoring

Microcell concept



- when the mobile station moves from one zone to next zone then the base station switches the channel to the new zone site. A given channel is active in only one zone. The channels are distributed in time & space are also reused in co-channel cells. This techniques particularly useful in highways or urban traffic corridors.

Advantages:-

- 1. The co-channel interference in the cellular system is reduced [illegible] BS is replaced by several low power

transmitters on the edge of the cell.

2. The signal quality is ↑ due to reduction in co-channel interference

3. channel capacity is ↑ without the degradation in trunking efficiency caused by sectoring.